

Early-20th-century US biologist Nettie Stevens made the pioneering discovery that an animal's sex is determined by particular chromosomes. She was one of the first female scientists to be recognized for her contribution to genetics, although her work was largely overshadowed during her lifetime.

MILESTONES

UNIVERSITY CAREER

Enrolls to study biology at Stanford University in 1896 following an early career in teaching.

EARNs DOCTORATE

Gains a PhD from Bryn Mawr in 1903 under the supervision of geneticist Thomas Hunt Morgan.

MEALWORM STUDY

Works on chromosomal behavior in mealworms sponsored by the Carnegie Institute in 1904.

MAJOR FINDING

Publishes her discovery of sex determination in 1905, just before US geneticist Edmund Beecher Wilson.

UNTIMELY END

Rejects an offer to be research professor at Bryn Mawr due to ill health and dies soon after in 1912.

Nettie Stevens graduated from Stanford University with a masters in biology in 1900, before gaining a doctorate at Bryn Mawr College in 1903. Although her scientific career did not begin until she was 39, Stevens is credited today for her vital breakthrough in early genetics.

Sex determination

In the early 20th century, scientists were divided over how biological sex was determined. Many believed it was caused by external factors at the embryonic stage, such as temperature or nutrition. Stevens's research into the chromosomal behavior of mealworms ended this debate.

During her study, she noted that the male reproductive cells included X and Y chromosomes, but the females produced only Xs. She therefore concluded that the sex of an organism is determined by the chromosomes it inherits from each parent and published her findings in 1905. This research made the first link between a physical characteristic and a particular chromosome. However, Stevens's discovery was not widely acknowledged until after her death.

“Her work will be remembered.”

Thomas Hunt Morgan, 1912

Stevens carried out painstaking work studying reproductive cells under a microscope, providing key data to support the theory of chromosomal inheritance. Her studies included many species of insects.

